

# **23MCT406**

# **ENVIRONMENTAL SCIENCE**

# MODULE 4 ~ Solid and Hazardous Waste Management

- Waste Management
- Consumerism and our throw-away culture
- Characteristics of municipal solid waste
- CPHEEO guidelines for solid waste management (overview only)
- Waste disposal methods (landfill, incineration, recycling)
- Sustainable practices in waste management ~ Transition to zero waste lifestyle – Circular Economy Hazardous and e-waste identification and management
- Recycling and waste-to energy technologies
- Regulations for hazardous waste management in India (overview only)
- Biomedical waste and its management

# SOLID WASTE

- Solid waste is the **unwanted or useless** solid materials
- Solid waste comprises of all the wastes arising from human and animal activities that are typically solid
- It is all-inclusive of the heterogeneous mass from the urban community as well as more homogeneous accumulation of agriculture and industrial wastes



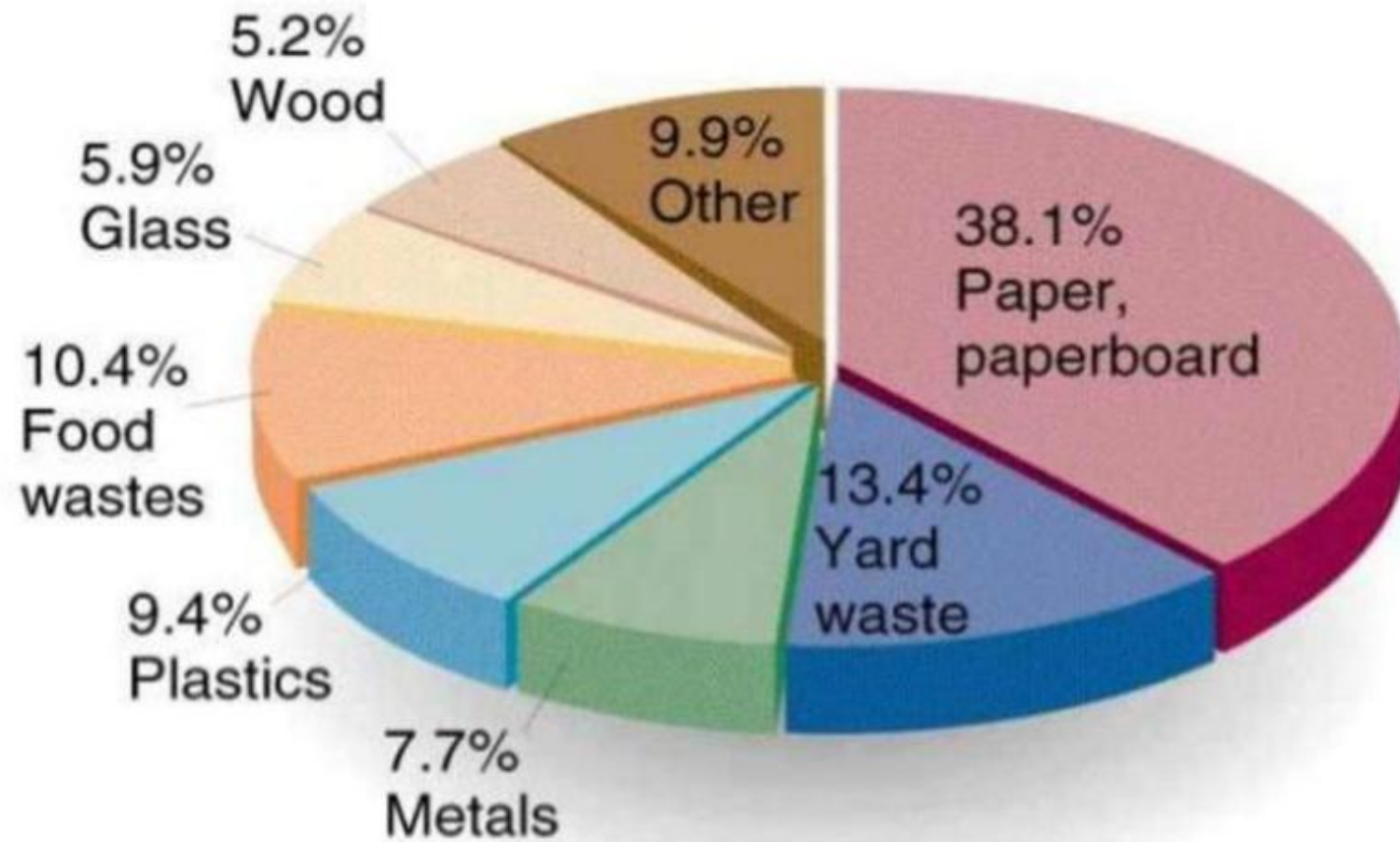
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- Solid wastes in general can be classified as **rubbish and garbage**
  - **Rubbish** refers to the solid waste components that will **not putrefy and that are either combustible** (such as paper, plastic, rubber etc.) or **non-combustible** (such as glass, metal, ceramics etc.)
  - **Garbage** refers to the solid waste components that **will putrefy** (such as fruits, fish, meat, vegetables etc.).

# Contd...

- It is estimated that in Indian cities on an average **200g to 600g** of solid waste is generated **per person per day**
- Solid waste includes:
  - ✓ Garbage (food wastes)
  - ✓ Rubbish (paper, plastics, wood, metal, throw away containers, glass)
  - ✓ Demolition products (bricks, masonry, pipes)
  - ✓ Sewage treatment residue
  - ✓ Dead animals, manure and other discarded material

## CHARECTERIZATION OF MUNCIPAL SOLID WASTE





# Environmental Impact of Solid Waste Disposal on Land

- Ground water contamination by the leachate generated by the waste dump.
- Surface water contamination by the run-off from the waste dump
- Bad odour, pests, rodents and wind-blown litter in and around the waste dump.
- generation of inflammable gas (e.g. methane) within the waste dump
- Bird menace above the waste dump which affects flight of aircraft.
- Fires within the waste dump
- Erosion and stability problems relating to slopes of the waste dump.
- Epidemics through stray animals
- Acidity to surrounding soil and release of green house gas.

# WASTE MANAGEMENT

Waste management is the process of:

- Collecting
- Treating
- Recycling
- Disposing

Different waste materials





# Consumerism and throw-away culture of solid waste

- Consumerism of solid waste refers to the direct link between excessive consumer habits and the generation of large amounts of solid waste
- A "throw-away culture" of solid waste is directly linked to consumerism
- To buy new products, often designed for single use, results in a significant increase in disposable waste
- This leads to overflowing landfills and environmental damage

## Contd...

- Excessive disposal of solid waste led to **reuse or recycle**
- People are encouraged to discard items quickly rather than repair or maintain them, contributing to a culture of waste generation

# Sources of Solid waste

## 1. Municipal Solid Waste (MSW)

Solid waste generated from

- Households (residential areas)
- Commercial sector (hotels and restaurants),
- Institutional buildings (offices and educational institutions) and
- Municipal sector (landscape and tree trimmings, construction and demolition waste and street sweepings)

## 2. Industrial Waste

- Solid waste generated from industrial activities
- Various industries (such as pulp and paper industry, chemical manufacturing industry etc.) generate a variety of wastes.
- Some are toxic (toxic chemicals)
- Some are recyclable (paper, plastic and metals)

# Contd...

## 3. Hospital waste

- Hospital waste is generated during the diagnosis treatment or immunization of human beings or animals
- It includes wastes the form of disposable syringes, needles, bandages, body fluids, contaminated organs of infected persons, etc. which are highly infectious
- Biomedical waste scattered in and around the hospitals invites flies, insects, rodent dogs, etc.
- They play an important role in spreading communicable diseases like plague and rabies

## 4. Hazardous waste

- Hazardous wastes are wastes that pose a substantial danger either immediately or over a period of time not only to living things (plants, animals and human beings) but also to the environment
- Typical examples are paints, radioactive wastes, pesticides, pathological wastes from hospitals etc
- They are acutely toxic which can cause cancer and trigger birth defects

# Contd...

## 5. Agricultural Waste

- Wastes generated from farms, dairies vineyards, fields etc.
- Mainly consist of spoiled food grains, vegetable wastes and animal wastes
- Animal and vegetable wastes can be used to increasing the soil fertility as they contain valuable nutrients

## 6. Plastic waste

- Plastic has become an intrinsic part of our lives at home, work and the market place
- This is the reason why we see plastic everywhere in our day-to-day life
- The lifestyle we follow today is "use and throw"
- It is estimated that approximately 56 lakhs tonnes of plastic waste is generated in India annually and most of it ends up on land and in waterways.

# Contd...

## **7. Construction and Demolition area**

Solid waste produced in these places include steel materials, concrete, wood, plastics, rubber, copper wires, dirt and glass

## **8. Municipal services**

Some of waste brought about by the municipal services include street cleaning, wastes from parks and beaches, waste water treatment plants, landscaping wastes and waste from recreational areas including sludge



# Contd...

## 9. E~ Waste

- E~ Waste covers any household or business item
- E~ waste contains circuits and electrical components (also known as electronic gadgets)
- Some of them are computers, mobile phones, air conditioners, televisions, etc.
- India ranks among the top 10 countries globally in terms of solid waste generation

# Waste Compositional Analysis

- Wastes are separated on the basis physical properties such as:

- Dry

- Wet

- Paper

- Glass

- Metal

- Household hazardous

- Construction and mixed/miscellaneous

# Contd...

- They are weighed and are represented as a fraction of total waste
- Importance of compositional analysis
  - ❖ Selection and operation of equipment and facilities
  - ❖ Design of disposal facility
- Variation of composition depends on:
  1. Location
  2. Season
  3. Economy

# Characteristics of municipal solid waste

1. Physical Characteristics
2. Chemical Characteristics
3. Biological Characteristics

# Physical Characteristics

## 1. Density

- The density of solid waste, i.e., its **mass per unit volume (kg/m<sup>3</sup>)**
- It is a critical factor in the design of an SWM system
- ❖ Example ~ design of sanitary landfills, storage, types of collection, and transport vehicles, etc.
- Usually, it refers to un-compacted waste
- Normal compaction equipment can achieve a **reduction in the volume of waste by 75%**
- It increases an initial density of 100 kg/m<sup>3</sup> to 400 kg/m<sup>3</sup>
- In other words, a waste collection vehicle can haul four times the weight of waste in its compacted state than when it is un-compacted

## 2. Moisture Content

- Moisture content is defined as the **ratio of the weight of water (wet weight - dry weight) to the total weight of the wet waste**
- It is one of the important physical characteristics of solid waste
- Moisture increases the weight of solid wastes, and thereby, the **cost of collection and transport**
- In addition, the moisture content is a critical determinant in the **economic feasibility** of waste treatment by incineration
- Wastes should be insulated from rainfall or other extraneous water
- A typical range of moisture content is 20% to 40%, representing the extremes of wastes in an arid climate and in the wet season of a region of high precipitation

### 3. PARTICLE SIZE

- It is an important characteristic of solid waste because of its significance in the design of mechanical separators and shredders
- The size and distribution of the components of wastes are also important for the recovery of materials
- Waste are separated by mechanical means such as trammel screens and magnetic separators
- For example, ferrous items which are of large size may be too heavy to be separated by a magnetic belt so magnetic separator is needed

## 4. Field capacity

- Field capacity of MSW is the total amount of **moisture that can be retained** in a waste sample subject to gravitational pull
- It is a critical physical characteristic of solid waste because the water in excess of field capacity will form leachate
- **Leachate can be a major problem in landfills**
- Field capacity varies with the degree of applied pressure and the state of decomposition of the wastes



# 5. Permeability

- Hydraulic conductivity of compacted waste
- It is an important physical property to a large extent govern movement of liquid and gases in landfill

# Chemical Characteristics

## 1. Volatile solid content

- Volatile matter exhibited an additional loss of weight on ignition at 950°C in a covered crucible

## 2. Ash content

- Ash content the amount of material remaining after burning the waste

OR

- Ash content exhibited a weight of residue after combustion in an open crucible

### 3.CHNSO content

- Carbon (C), hydrogen (H), nitrogen (N), sulphur (S), and oxygen (O) content
- The C, H, N, S and O contents were measured via an elemental analyzer (CE Instruments, CHNS-O Model EA-1110, Thermo Electron)

#### 4. Calorific value

- An evaluation of the potential of waste material for use as fuel for incineration requires a determination of its heating value expressed as kilojoules per kilogram (kJ/kg).
- The calorific value or energy content of solid waste was determined by using a laboratory bomb calorimeter

## Ultimate analysis:

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This refers to an analysis of waste to determine the proportion of **carbon, hydrogen, oxygen, nitrogen and sulphur** and the analysis is done to make mass balance calculation for a chemical or thermal process.

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Besides, it is necessary to determine **ash fraction** because of its potentially harmful environmental effects, brought about by the presence of toxic metals such as cadmium, chromium, mercury, nickel, lead, tin and zinc.

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Note that other metals (e.g., iron, magnesium, etc.) may also be present but they are non-toxic.

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The results are used to **characterize the chemical composition of the organic matter** in MSW.

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Used to define proper mix of waste material to achieve **suitable C/N ratios** for **biological conversion processes**.

## Proximate analysis:

Proximate analysis of waste aims to determine moisture, volatile matter, ash and fixed carbon. This is important in evaluating the combustion properties of wastes or refuse derived fuel.

The fractions of interest are:

- moisture content which adds weight to the waste without increasing its heating value, and the evaporation of water reduces the heat released from the fuel
- ash which adds weight without generating any heat during combustion
- volatile matter i.e. that portion of the waste that is converted to gases before and during combustion
- fixed carbon which represents the carbon remaining on the surface grates as charcoal. A waste or fuel with a high proportion of fixed carbon requires a longer retention time on the furnace grates to achieve complete combustion than a waste or fuel with a low proportion of fixed carbon

# Biological Characteristics

- The most important biological characteristic of the organic fraction of solid waste is that can be converted biologically to gases and relatively inert organic and inorganic solid
- generation of flies are also related to the putrescible nature of the organic materials.

## Volatile Solids

- Volatile solids content is often used as a measure of the biodegradability of the organ
- It is not an exact indicator of biodegradability
- Some organic constituents are highly volatile but low in biodegradability



# CPHEEO guidelines for solid waste management

# Waste disposal methods

1. Landfill
2. Incineration
3. Recycling

# 1. Landfill

- Landfill is the site for the disposal of waste materials **by burial**
- **It is the oldest form of waste treatment**
- Modern sanitary landfills are carefully engineered structures designed to separate garbage from nearby water, soil, wildlife, and people
- Cavities are dug out of the ground
- Walls are sealed with **layers of clay and coated with plastic** to prevent groundwater contamination from wastewater that accumulates



Open space dumping



Burial

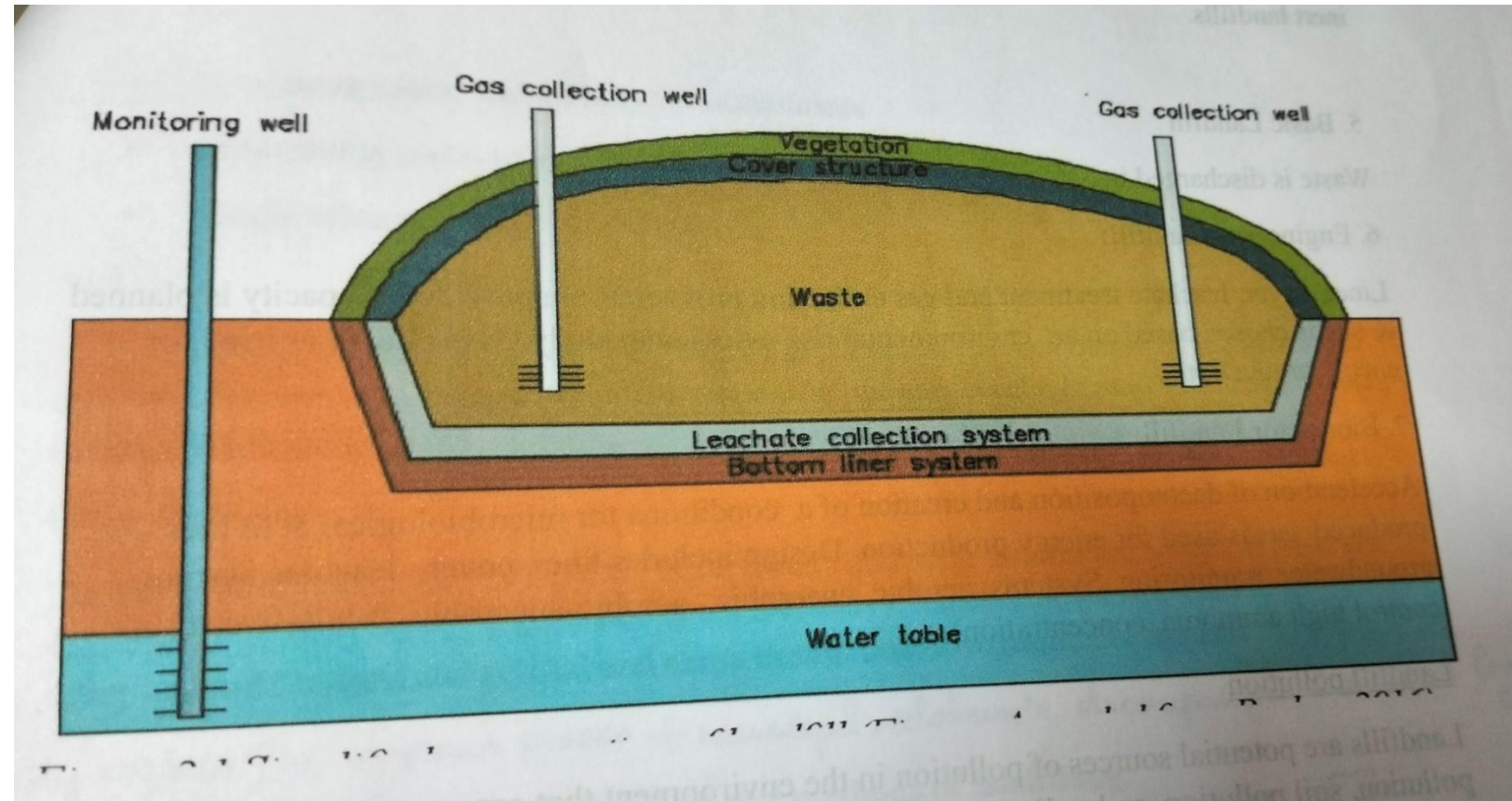
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- Any waste that is not recycled or reused has to go somewhere ends up at a landfill site
- Landfill sites can be created from a void made by quarrying
- It contain both household and commercial rubbish
- For example, food, paper, cardboard or wood
- It also contain wastes from construction and commercial businesses, such as soil, concrete and brick rubble

# Contd...

Main components of landfill are:

1. Bottom liner
2. Cells
3. Leachate collection system
4. Storm water drainage
5. Methane collection system
6. Cover
7. Groundwater monitoring stations



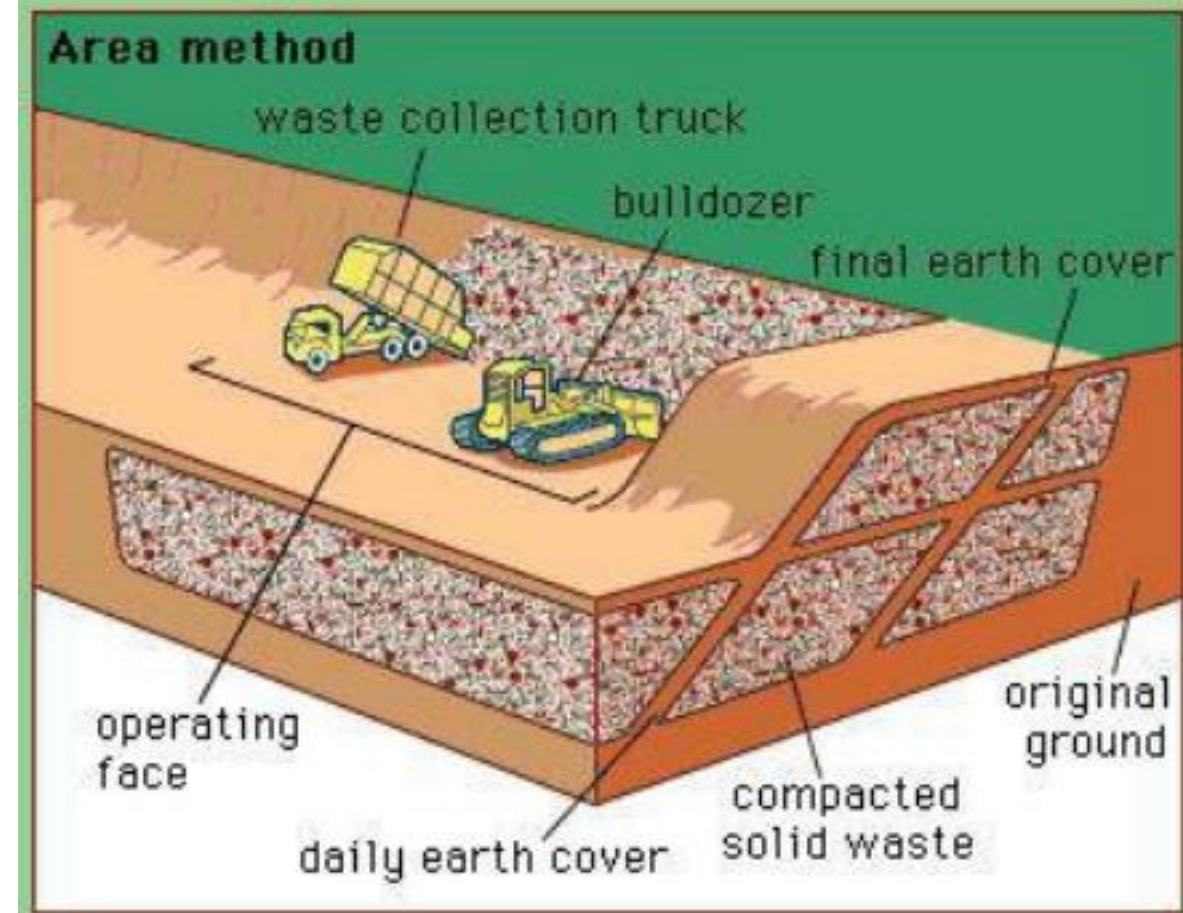
# Methods of Landfilling

- a) Area method
- b) Trench method
- c) Ramp/slope method
- d) Valley method
- e) Cell method
- f) Combination method



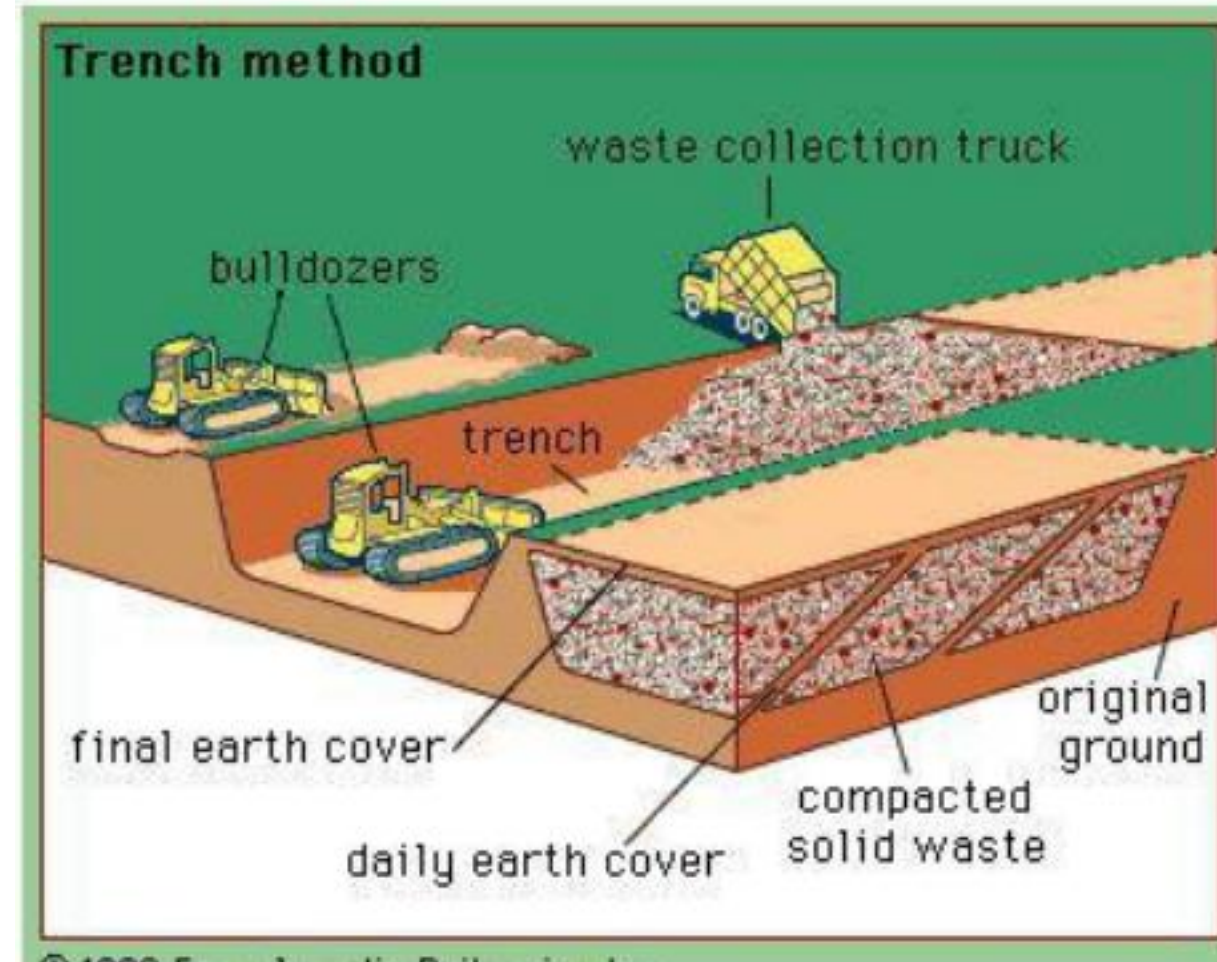
## a) Area method

- Area landfilling method is also known as **Above Ground Landfill**
- Area landfilling is mostly used to dispose urban waste
- This method is best suited for flat or gently sloping areas where some land depressions may exist



## b) Trench method

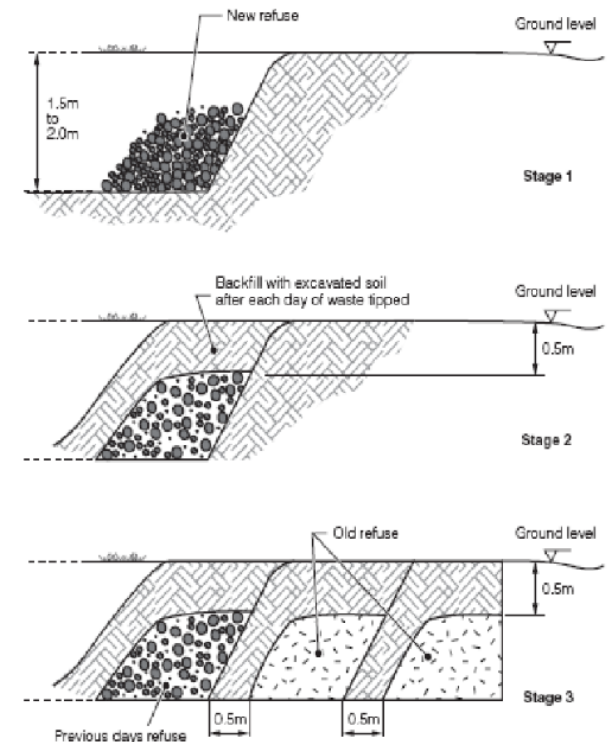
- Trench method is also known as **Below Ground Landfill**
- Trench method of landfilling involves excavation of a waste trench
- Collected waste is disposed inside excavated trench and is covered-up with a layer of soil
- Method is suitable only for **small quantity of waste**





## c) Ramp/slope method

- Involves disposing collected waste along the sides of **existing hill slope**
- The slope or ramp is sometimes used in combination with the other two methods
- Wastes are spread on an existing slope, compacted and covered
- This variation may be suitable for most areas



## d) Valley method

- Valley Landfilling Method is also known as **Canyon Landfills/depression method**
- In valley landfilling method collected waste is deposited in valleys, low-lying areas, canyons, ravines etc.
- Used where natural or artificial depressions exist and are used for land filling
- Method depends on the geometry of the site, hydrology and geology of the site, and access to site

## e) Cell method

- It is a progressive filling and restoration concept
- Method advocates tidy disposal of waste and involves depositing collected waste in a pre-constructed bonded area

## f) Combination method

- More than one of the above listed landfilling methods are employed in combination to dispose off collected waste

# Advantages of landfilling

- Simple and economical. No costly plant or equipment is required
- Separation of different kinds of refuse is not required
- No residues are evolved and hence no further disposal is required
- Low-lying water-logged areas and odd quarry pits can be easily recovered and put to better use
- Mosquito breeding places are also thus eliminated

# Disadvantages of sanitary landfilling

- Low lying depressions or dumping sites may not always be available
- Evolution of landfill gases
- Highly toxic and polluted leachate generation

## 2. Incineration

- Method also known as combustion
- Waste is burnt at high temperatures to convert into gaseous compounds
- It reduces volume of waste by 20~30% of original volume
- This process is also known as thermal treatment where solid waste materials are converted by Incinerators into heat, gas, steam and ash.

# Pyrolysis

- Heating in closed containers in oxygen free atmosphere, most of the organic substances of solid waste can be split through a combination of thermal cracking and condensation reactions in to gaseous, liquid and solid fractions
- It is highly endothermic

# Contd...

## ❖ INCINERATORS

- An apparatus for burning waste material, especially industrial waste, at high temperatures until it is reduced to ash
- It reduces the volume of the waste by removing combustible matter





# Contd...

Different types of incinerators include:

- Fluidized bed
- Injection incinerator
- Multiple hearth incinerator
- Waste gas flare
- Direct flame

# Advantages of incineration

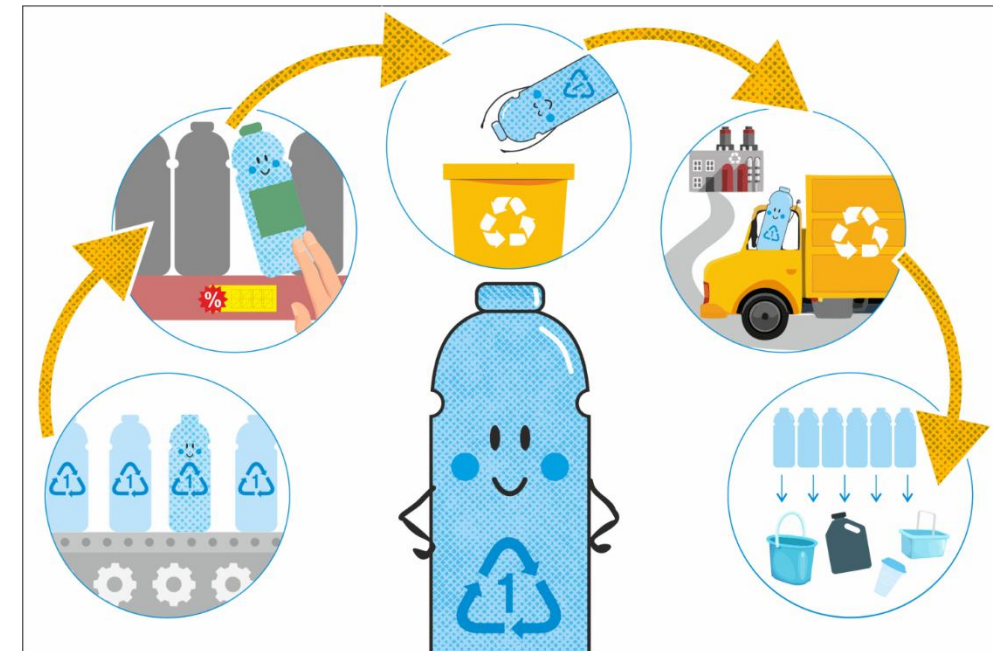
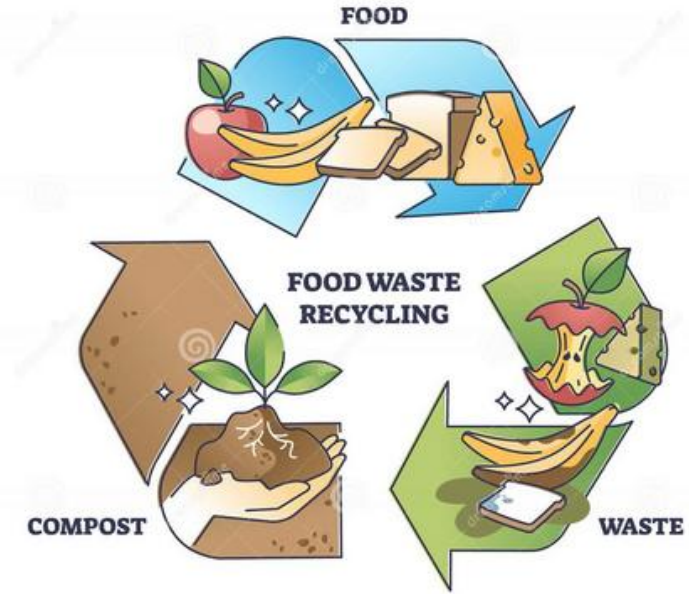
- Decreases quantity of waste
- Efficient waste management
- Production of heat and power
- Reduction of pollution
- Incinerators have filters for trapping pollutants
- Saves on transportation of waste
- Provides better control over odour and noise
- Prevents the production of methane gas
- Eliminates harmful germs and chemicals
- Incinerators operate in any weather
- It has a computerized monitoring system
- Occupies relatively small space
- Uncontaminated groundwater

# Disadvantages of incineration

- Expensive
- Pollutes the environment
- Damaging public health
- The possibility of long-term problems
- Ash waste can potentially harm people and the environment

### 3. Recycling

- Recycling is a process of salvaging valuable materials from the waste generated so that it can be used again as the same material.
- Eg: Recycling of food wastes, Recycling of plastics, glass, etc.
- Recycling can be a primary or a secondary process



# Contd...

- Primary recycling is process in which the original waste material is made back into the same material
- In secondary recycling waste materials are made into different products that may or may not be recyclable
- For eg:
  - Cardboards made from waste news paper
  - Shredded used tires used in construction of pavements
  - compost from food wastes

# Sustainable waste management system

- Sustainable waste management system is an efficient way to minimize the hazards or pollution caused by wastes
- Sustainable waste management means using material resources efficiently to cut down the number of wastes produced and where waste is generated, dealing with it in a way that actively contributes to the economic, social and environmental goals of sustainable development

# Sustainable practices in waste management

Sustainable waste management practices include

- Reducing waste
- Reusing materials
- Recycling
- Composting

# 3R CONCEPTS

- Reduce, Reuse and Recycle (3R) are the three essential components of environmentally-responsible consumer behavior





# REDUCE

- Reduce Use things with care
- Buy fewer things
- Avoid single-use items

# RECYCLING

- **Recycle** means to convert waste into material that can be used to remake the item, or to make something else.
- Series of steps that take a used material and processes, remanufactures and sell it as a new product
- Recycling is the processes of converting waste materials in to new materials and objects
- Buy product made from recycled material
- Consider purchasing recycled materials for office supply, office equipment or manufacturing

# REUSE

- Reuse means to use the same item more than once, preferably many times rather than disposing of it in one use
- Economic and environmental sense to reuse products
- Sell old cloths, appliances, toys or donate them to charities
- Use ceramic coffee mug instead of paper cups



# Waste hierarchy

Waste hierarchy refers to 3 Rs  
Reduce, Reuse, Recycle



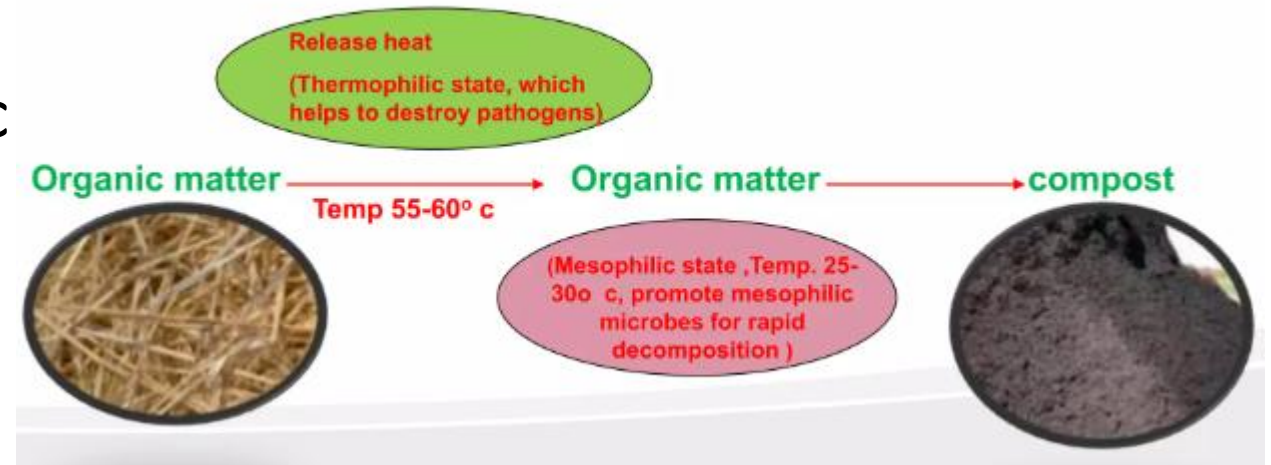
# COMPOSTING

- Biological method for decomposing solid waste
- Work in aerobic and anaerobic face
- Final product – compost or humus– use as manure



Indore method - Aerobic

Bangalore composting- Anaerobic





## Benefits of Compost

Compost improves the quality of soil, and for this reason it is considered as a soil conditioner.

It contains a variety of the basic nutrients required for healthy growth of plants.

In addition to, nitrogen, phosphorous, and potassium, certain micronutrients *viz.* manganese, copper, iron, and zinc also found in compost which helps them to control diseases and insects.

Compost improves the structure and texture of the soil enable them to retain nutrients, moisture, and air for the betterment of growth of plants.





## Vermicomposting



- Worm composting is using worms to recycle food scraps and other organic material into a valuable soil amendment called vermicompost.
- It is a mesophilic process, utilizing microorganisms and earthworms that are active at 10–32°C.
- The process is faster than composting; because the material passes through the earthworm gut, whereby the resulting earthworm castings ie, rich in nutrients.
- Earthworms consume various organic wastes and reduce the volume by 40–60%.

# WHAT SHOULD BE DONE

- **Reduce Waste**

- Reduce office paper waste by implementing a formal policy to duplex all draft reports and by making training manuals and personnel information available electronically.
- Improve product design to use less materials.
- Redesign packaging to eliminate excess material while maintaining strength.
- Work with customers to design and implement a packaging return program.
- Switch to reusable transport containers.
- Purchase products in bulk.



# WHAT SHOULD BE DONE

## Reuse

- Reuse corrugated moving boxes internally.
- Reuse office furniture and supplies, such as interoffice envelopes, file folders, and paper.
- Use durable towels, tablecloths, napkins, dishes, cups, and glasses.
- Use incoming packaging materials for outgoing shipments.
- Encourage employees to reuse office materials rather than purchase new ones.

# WHAT SHOULD BE DONE

## Donate/Exchange

- old books
- old clothes
- old computers
- excess building materials
- old equipment to local organizations

# WHAT SHOULD BE DONE

## Employee Education

- Develop an “office recycling procedures” packet.
- Send out recycling reminders to all employees including environmental articles.
- Train employees on recycling practices prior to implementing recycling programs.
- Conduct an ongoing training process as new technologies are introduced and new employees join the institution.

# WHAT SHOULD BE DONE

## Employee Education

- education campaign on waste management that includes an extensive internal web site, quarterly newsletters, daily bulletins, promotional signs and helpful reference labels within the campus of an institution.

# WHAT SHOULD BE DONE

**Conduct outreach program adopting an ecologically sound waste management system which includes:**

- waste reduction
- segregation at source
- composting
- recycling and re-use
- more efficient collection
- more environmentally sound disposal

Residents may be organized into small groups to carry out the following:

1. **construction of backyard compost pit**
2. **construction of storage bins where recyclable and reusable materials are stored by each household**
3. **construction of storage centers where recyclable and reusable materials collected by the street sweepers are stored prior to selling to junk dealers**
4. **maintenance of cleanliness in yards and streets**
5. **greening of their respective areas**
6. **encouraging others to join**